Daffodil

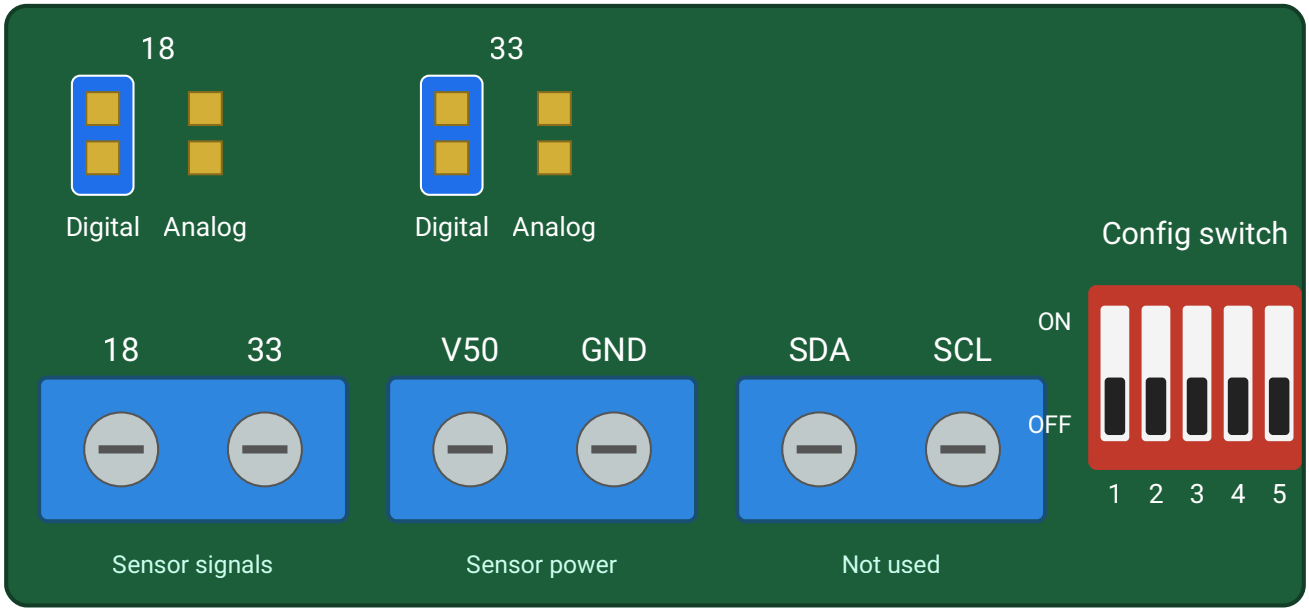
User Manual: setting up and reading your sensor unit

Daffodil watches water for you. It reads water troughs, tanks, septic tanks, water flow and water temperature, then reports to Digital Stables over WiFi and to nearby units over LoRa radio. It runs on solar power and a battery.

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1. Getting to know your Daffodil

Everything you need to set up is on the lower edge of the board:



Part	What it does
18 / 33 terminal	The signal wires of your sensors go here. Terminal 18 is sensor 1 and terminal 33 is sensor 2.
V50 / GND terminal	Powers your sensors. Every sensor's + wire goes to V50 and its − wire goes to GND. When two sensors are connected, both + wires share V50 and both − wires share GND.
18 and 33 jumpers (marked "Digital Analog")	Each sensor terminal has two small 2-pin headers, one labelled Digital and one labelled Analog. Put one jumper cap on the header your sensor needs and leave the other one empty. The drawing above shows both caps on Digital.
Config switch (1–5)	Tells Daffodil what it is measuring (switches 1–4) and how often to report (switch 5). Up is ON.
SDA / SCL terminal	Not needed for any of the setups in this manual.

Which header? Use `Digital` for flow meters, ultrasonic level sensors and the water-temperature probe. Use `Analog` only for tank pressure sensors. Never put a cap on both headers of the same terminal.

Always switch the unit off (unplug USB and the battery) before changing wires, jumpers or switches. Daffodil only reads the config switch when it starts up.

2. Quick setup table

Find what you want to measure, then set the switches, sensors and jumper caps exactly as shown. In the switch pictures, a black knob at the top means ON. Switch 5, shown with a ?, is your choice (see section 3).

Switches 1–4	Setting	Terminal 18	18 jumper	Terminal 33	33 jumper
 off off off off	1 Flow Sensor Measures water flow through one pipe.	Flow meter	 Digital	nothing	 No cap
 ON off off off	2 Flow Sensors Measures flow through two pipes.	Flow meter 1	 Digital	Flow meter 2	 Digital
 off ON off off	1 Flow Sensor + 1 Tank One pipe's flow plus one tank's level.	Flow meter	 Digital	Tank pressure sensor	 Analog
 ON ON off off	1 Tank Level of one tank, using a pressure sensor at the bottom.	Tank pressure sensor	 Analog	nothing	 No cap
 off off ON off	2 Tanks Levels of two tanks.	Tank 1 pressure sensor	 Analog	Tank 2 pressure sensor	 Analog
 ON off ON off	Septic Tank How full a septic tank is, measured from above.	Ultrasonic sensor	 Digital	nothing	 No cap
 off ON ON off	Water Trough Water level in a trough, measured from above.	Ultrasonic sensor	 Digital	nothing	 No cap
 ON ON ON off	Water Trough + Tank A trough level plus one tank level. The ultrasonic goes on terminal 33 in this setting.	Tank pressure sensor	 Analog	Ultrasonic sensor	 Digital
 off off ON ON	2 Water Troughs Water levels in two troughs. Both levels are reported, but the lights don't show them yet.	Ultrasonic sensor 1	 Digital	Ultrasonic sensor 2	 Digital
 off off off ON	Water Trough + Water Temperature A trough level plus the water temperature.	Ultrasonic sensor	 Digital	Water temperature probe	 Digital

Jumper pictures: D = Digital header, A = Analog header. The blue block shows where the cap goes. Any switch combination not in this table is unused, and the unit won't measure anything in it.

3. Switch 5: how often to report

Switch 5 ON: Battery Saver

Daffodil plans its day around the sun. It sleeps longer and reports less often, so the battery lasts as long as possible. Best for most installations, such as a trough level that only needs checking a few times an hour.

Switch 5 OFF: Frequent Reports

Daffodil reports more often and doesn't wait for good sun. You get readings closer to real time, but it uses more battery. Best when you need to watch something closely, such as water flow, or when the unit has reliable power.

Either way, Daffodil protects its own battery. If the battery gets low it turns off its lights first, then WiFi, then goes to sleep until the sun has recharged it.

At night Daffodil still reports at least every 5 minutes while its battery is healthy, and every 90 seconds in the hour and a half before sunrise, to catch the coldest part of the night. If the battery runs low it sleeps longer between reports to save it.

4. Step-by-step setup

1. Unplug the battery and USB.
2. Set switches 1–4 for what you're measuring (section 2), and switch 5 for how often it reports.
3. Fit the jumper caps for terminals 18 and 33 as shown in section 2.
4. Connect each sensor: its signal wire to terminal 18 or 33, its + wire to V50 and its − wire to GND (section 5).
5. Connect the battery (and USB or solar panel). The lights start cycling through their displays.
6. For trough and septic setups, enter the sensor's mounting height (section 6).
7. Watch the lights for a minute and check the level, internet and LoRa displays look right (section 7).

The setups one by one:



1 Flow Sensor

Measures water flow through one pipe.

- Terminal 18: Flow meter signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



2 Flow Sensors

Measures flow through two pipes.

- Terminal 18: Flow meter 1 signal wire. Cap on the 18 Digital header.
- Terminal 33: Flow meter 2 signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).



1 Flow Sensor + 1 Tank

One pipe's flow plus one tank's level.

- Terminal 18: Flow meter signal wire. Cap on the 18 Digital header.
- Terminal 33: Tank pressure sensor signal wire. Cap on the 33 Analog header.
- Every sensor's power wires: V50 (+) and GND (−).



1 Tank

Level of one tank, using a pressure sensor at the bottom.

- Terminal 18: Tank pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



2 Tanks

Levels of two tanks.

- Terminal 18: Tank 1 pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: Tank 2 pressure sensor signal wire. Cap on the 33 Analog header.
- Every sensor's power wires: V50 (+) and GND (−).



Septic Tank

How full a septic tank is, measured from above.

- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



Water Trough

Water level in a trough, measured from above.

- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



Water Trough + Tank

A trough level plus one tank level.

- Terminal 18: Tank pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: Ultrasonic sensor signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

The ultrasonic goes on terminal 33 in this setting.



2 Water Troughs

Water levels in two troughs.

- Terminal 18: Ultrasonic sensor 1 signal wire. Cap on the 18 Digital header.
- Terminal 33: Ultrasonic sensor 2 signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

Both levels are reported, but the lights don't show them yet.



Water Trough + Water Temperature

A trough level plus the water temperature.

- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: Water temperature probe signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

5. Connecting the sensors

Ultrasonic level sensor (troughs and septic tanks)

Use a waterproof ultrasonic sensor with automatic serial (UART) output, such as the A02YYUW. JSN-SR04T and AJ-SR04M sensors also work if set to their automatic serial mode. The older style of sensor with separate "trigger" and "echo" wires and open speakers does not work with current Daffodils. Those sensors corroded in the field.

Sensor wire	Connect to
VCC / + (usually red)	V50
GND / − (usually black)	GND
TX	Terminal 18. In Water Trough + Tank, terminal 33. In 2 Water Troughs, the second sensor goes on 33.
RX	Not connected. Tape it off.

Jumper cap on that terminal's Digital header.

> Mounting: point the sensor straight down at the water, clear of the trough walls, float valves and pipes. Keep its face at least 3 cm above the highest water level, because it can't measure anything closer than that.

Water temperature probe

A waterproof DS18B20 probe, the common stainless-steel tube on a cable. It is used only in the Water Trough + Water Temperature setting.

Probe wire	Connect to
Red	V50
Black	GND
Yellow (data)	Terminal 33, jumper cap on the 33 Digital header

No extra resistor is needed, because the board already has one. Put the probe tip in the water, away from the trough's inlet, so it measures the water the animals are actually drinking.

Flow meter

A standard 3-wire hall-effect water flow meter.

Meter wire	Connect to
Red	V50
Black	GND
Yellow (pulse)	Terminal 18 (flow meter 1) or 33 (flow meter 2), cap on Digital

Install the meter with the arrow on its body pointing in the direction the water flows.

Tank pressure sensor

A submersible or bottom-mounted pressure sensor with a 0.5–4.5 V output (0–5 psi), placed at the bottom of the tank.

Sensor wire	Connect to
+ (usually red)	V50
− (usually black)	GND
Signal	Terminal 18 (tank 1) or 33 (tank 2), cap on Analog

6. Setting up a trough or septic tank

For troughs, Daffodil needs to know how high the sensor is and where you want the "low" and "full" points. Open the Daffodil's web page, tap the gear icon on the level card, and enter:

Setting	What to enter
Sensor height	Distance in cm from the sensor's face down to the empty trough floor.
Level minimum	Water depth in cm below which the trough counts as low (red light).
Level maximum	Water depth in cm above which the trough counts as full (blue light).

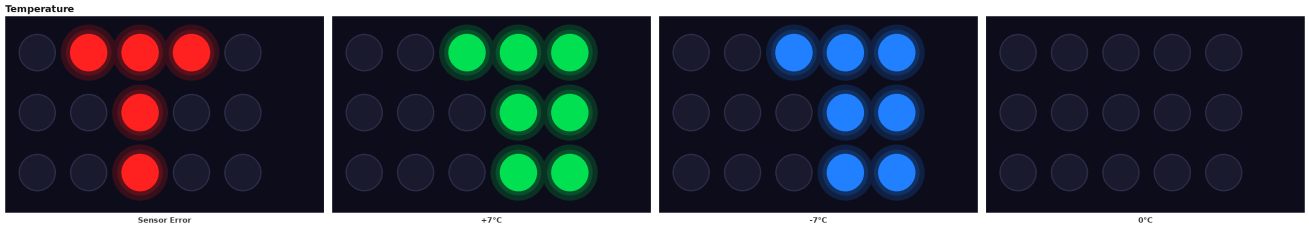
Example: the sensor is 60 cm above the trough floor, you want an alert below 20 cm of water, and you call it full above 30 cm. Enter 60, 20 and 30. Re-enter these whenever you move the sensor or change the trough.

A septic tank needs no settings. Its lights show how full the tank is, measured from a sensor mounted at the top.

7. Reading the lights

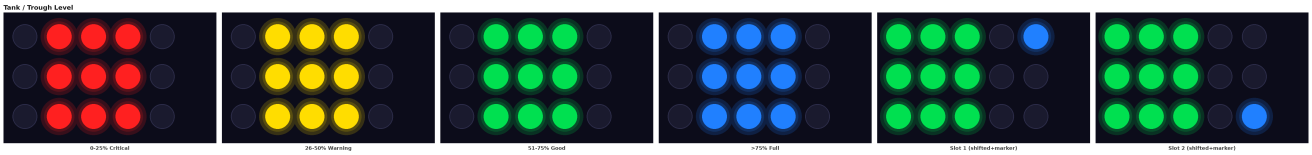
Daffodil has 15 lights in 3 rows of 5. Every few seconds the display moves on to show the next piece of information, in this order.

Temperature



The air temperature. The left lights count the tens and the right lights count the units. Green means above zero, blue means below zero, and yellow means exactly 0 °C. A red pattern means the temperature sensor couldn't be read.

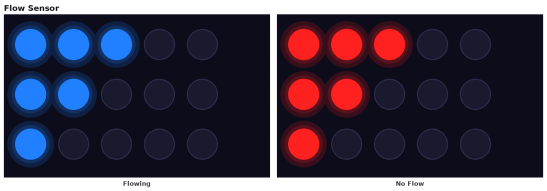
Water level (troughs, tanks, septic)



Red = low or critical, yellow = getting low (tanks and septic only), green = good, blue = full. For troughs, red, green and blue follow the levels you set in section 6. With two sensors, the block shifts left and a small blue dot shows which one you're seeing: top-right for sensor 1, bottom-right for sensor 2.

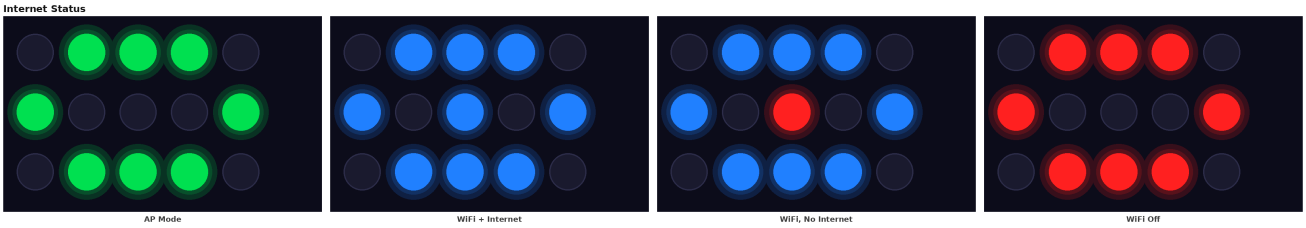
> A trough that always shows blue may mean the sensor isn't being heard at all. Check its wiring (section 9).

Water flow



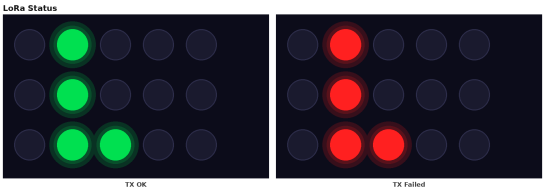
An "F" shape. Blue = water is flowing, red = no flow right now.

Internet



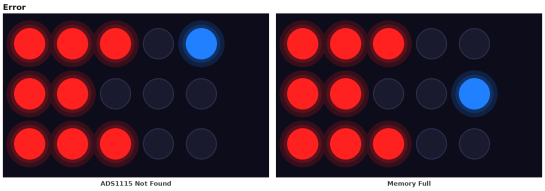
Blue = connected to your WiFi. The centre light is blue when the internet can be reached and red when it can't. Green = setup mode: Daffodil is broadcasting its own WiFi network for you to connect to. All red = WiFi is resting to save battery.

LoRa radio



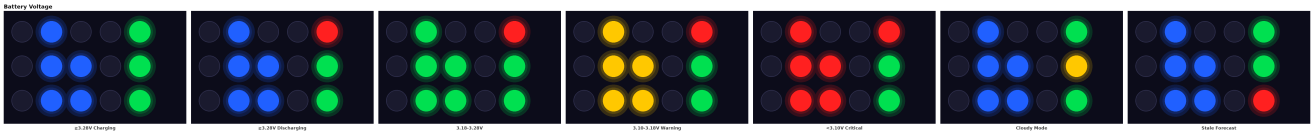
Green = the last radio message went out, red = it failed. The lights blink off briefly each time a message is sent. That's normal.

Error



A red "E" with a blue dot. A dot top-right means an internal sensor wasn't found. A dot in the middle-right means the unit's memory is nearly full. Either way, contact Digital Stables.

Battery



The "B" shows battery health: blue = good, green = OK, amber = low, red = very low, magenta = no battery detected. The top-right light is green while charging and red while running on battery. The middle-right light is amber on cloudy days, when the unit is saving power.

8. Looking after your Daffodil

- Keep the solar panel clean and in full sun.
- Every few months, check the ultrasonic sensor's face is clean and the cable glands are tight.
- If you change the battery, the config switch may need recalibrating. Use Config Switch Calibration on the web page, or ask Digital Stables.
- If you move a sensor or change a trough, re-enter its heights (section 6).
- After a firmware update over USB, move the jumper from the USB position back to the watchdog header. Left on USB, the USB chip stays powered and can flatten the battery overnight. Annabelle's screen then shows !! USB JUMPER ON !!, and on the Teleonome page the unit's name turns red with USB header connected.

9. Troubleshooting

Problem	What to check
The level never changes, or a trough always shows blue	The ultrasonic's TX wire is on the right terminal (section 2) and its cap is on Digital. It has power from V50/GND, and it's an automatic serial sensor.
Water temperature missing	The probe's yellow wire is on terminal 33 with the cap on 33 Digital, and switches 1–4 are set to Water Trough + Water Temperature.
Flow always shows red (no flow)	The meter's yellow wire is on the right terminal with its cap on Digital, and the meter's arrow points the way the water flows.
Tank level wrong or stuck	The cap for that terminal is on Analog, not Digital.
The unit measures the wrong thing	Switches 1–4 are set correctly and the unit was restarted after changing them. If they are, the switch needs recalibrating (section 8).
Internet lights all red	Normal when the battery is low or it's very cloudy. Daffodil rests WiFi to save power and tries again later.
The unit's name is red with USB header connected	The programming jumper was left on the USB position after an update. Move it back to the watchdog header (section 8).
An orange or red Reset … ago tag on the Teleonome page	The unit restarted. Orange (power-on) is normal after an update, a replug or a battery change. Red (brownout, crash or watchdog) means something went wrong: if it keeps coming back, contact Digital Stables.

A % delivered tag, or readings arriving with long gaps	Some radio messages are not reaching Annabelle. Check the antenna is screwed on and upright, and that nothing new (a shed, a vehicle, a metal tank) stands between the unit and Annabelle.
No lights, or very dim lights	Normal at night (the lights dim), when the battery is low, and on dull days in Battery Saver mode (the lights turn off to save power). Wait for sun, or check the battery is connected.

Still stuck? Contact Digital Stables with the unit's name and a description of what the lights are doing.